To ensure a scientifically rigorous comparison between an LLM and an SLM—especially in the complex domain of multimodal video contextualization—several variables must be controlled. Comparing a 70B parameter model to a 7B parameter model is inherently asymmetric, so this evaluation framework needs to account for differences in hardware, input processing, and evaluation metrics to prevent skewed results.

Following is a breakdown of the key biases with their mitigation strategies:

1. Input and Context Bias (The "Field of View")

Massive models often have much larger context windows (e.g., Gemini 1.5 Pro's 2 million tokens vs. an SLM's 8k to 32k tokens).

- **The Bias:** If you allow the LLM to process 100 frames of a video and only allow the SLM to process 16 frames due to its context limit, the LLM will naturally win because it "saw" more of the data, not necessarily because its architecture is better at reasoning.
- **The Mitigation:** Standardize the input. Force both models to process the exact same number of frames, sampled at the exact same timestamps, using the exact same resolution. This tests their reasoning capability over identical data.

2. The Vision Encoder Discrepancy

In multimodal models, the "brain" (the language parameters) is separate from the "eyes" (the vision encoder, like CLIP or SigLIP).

- **The Bias:** An 8B parameter SLM might beat a 70B parameter LLM in video understanding simply because the SLM is paired with a newer, higher-resolution vision encoder. The comparison becomes a test of the encoders, not the models themselves.
- **The Mitigation:** Explicitly document the vision encoders used by each model. When analyzing the results, isolate whether failures were due to "visual blindness" (the encoder missed it) or "reasoning failure" (the text model misunderstood it).

3. Prompting and Instruction-Tuning Bias

Massive closed-source models have undergone millions of hours of RLHF (Reinforcement Learning from Human Feedback), making them incredibly good at understanding complex, zero-shot prompts.

- **The Bias:** If you use a highly complex, multi-step prompt, an SLM might fail simply because it struggles with instruction-following formatting, even if it actually understood the video content.
- **The Mitigation:** Use **Few-Shot Prompting** for both models, or explicitly tailor the prompts to match the format each model was trained on. Furthermore, use structured outputs (like forcing both models to output strict JSON) so you aren't penalizing the SLM for having a less conversational tone.

4. Expanding the Evaluation Metrics (Beyond Accuracy)

If you only measure "Accuracy," you inherently bias the results toward the LLM. A fair comparison must measure the *cost* of that accuracy.

- **The Bias:** An LLM might score 75% while the SLM scores 70%, making the LLM look like the clear winner if accuracy is the only metric.
- **The Mitigation:** Implement a multidimensional evaluation matrix. You must measure:
  - **Task Accuracy:** (e.g., exactly matching the correct multiple-choice answer on Video-MME).
  - **Inference Latency:** Time to First Token (TTFT) and total generation time.
  - **Compute Efficiency:** Tokens generated per second (TPS).
  - **Memory Footprint:** Peak VRAM required during inference.

5. World Knowledge vs. Grounded Reasoning

LLMs have memorized a massive portion of the internet.

- **The Bias:** In video question-answering tasks, an LLM might guess the right answer based on its text-based world knowledge without actually looking at the video, artificially inflating its score.
- **The Mitigation:** Evaluate using datasets that require strict **temporal grounding** (e.g., "At what exact timestamp did the car turn left?"). This prevents the LLM from relying on its generalized internet knowledge and forces both models to prove they are actively contextualizing the video frames in front of them.